



reinforcement: $y_{r,t+1} = (1 - \rho)y_{r,t} + \rho \tilde{y}_r$

corrective rate: $y_{r,t+1} = y_{r,t} + \kappa(\pi_{type(r)} - y_{r,t}) + \epsilon$

associative strength update:
 $P_{r,t+1} = \text{logit}^{-1}(\text{logit}(P_{r,t}) + \Delta \times f(O_{r,t}))$

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success: $O_{r,t} \sim \text{Bernoulli}(P(r)) = 1$

failure: $O_{r,t} \sim \text{Bernoulli}(P(r)) = 0$

ϵ lexical likelihood: $\exp(-(\tilde{x}_r - \mu_l)^2 / 2\sigma_x^2)$
 λ iconicity evidence: $E(\tilde{y}, \pi_{type}) = 2 \exp(-k|\tilde{y}_r - \pi_{type}|) - 1$
 ω combined logits: $\text{logit}_r = \log(\text{lexical likelihood}) + \text{logit}(P_{r,t}) + \omega E(\tilde{y}_r, \pi_{type})$
choice probability: $P(r) = \text{softmax}(\lambda \times \text{logit}_r)$

lexical dimension: $\tilde{x} \sim \mathcal{N}(x_r, \sigma_x^2)$

iconic dimension: $\tilde{y} \sim \mathcal{N}((1 - \alpha)y_r + \alpha\pi_{type(r)}, \sigma_y^2)$

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